

John Smith, Jane Doe
Journal of Shoulder and Elbow Surgery, 2024

Abstract

This study compares the biomechanical properties of various suture anchor configurations used in rotator cuff repair surgery.

Introduction

Overview of suture anchor placement techniques.

Rotator cuff tears are a common shoulder pathology. Surgical repair using suture anchors remains the gold standard treatment.

Methods

Patient Demographics

Table 1: Patient demographics and baseline characteristics (n=45).

Biomechanical Testing

Figure 2: Biomechanical testing setup. (A) Custom loading fixture. (B) Cyclic loading protocol.

Results

Load to Failure

Figure 3: Load to failure comparison. Error bars represent standard deviation. *p < 0.05 vs single row.

Stiffness Analysis

Figure 4: Stiffness comparison between repair techniques.

Table 2: Biomechanical properties summary.

Discussion

The double row technique demonstrated significantly higher load to failure compared to single row repair.

Conclusion

Double row repair provides superior biomechanical properties compared to single row repair.

Double row suture anchor fixation provides superior biomechanical properties for rotator cuff repair.